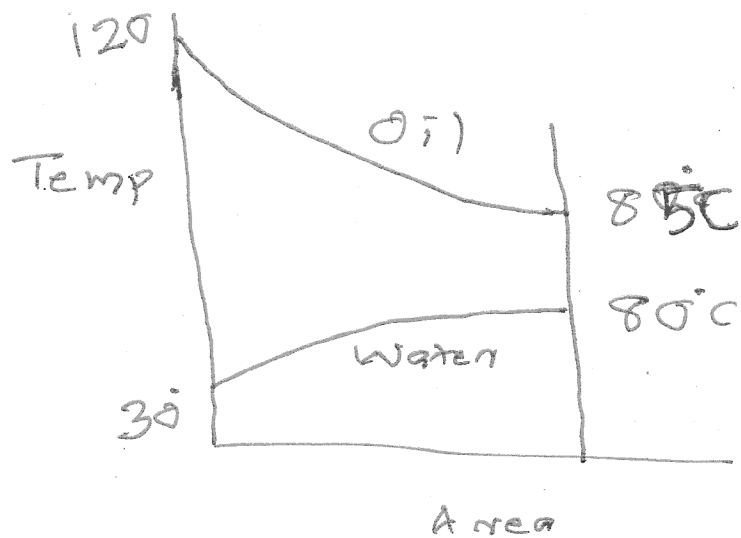


① Tutorial
Qn-2.



Hot fluid - Oil, $m_h = 9$
 $c_{ph} = 9$

Cold fluid Water $m_c = 0.8 \text{ kg/s}$

$c_{pc} = 4.18 \text{ kJ/kgK}$

(from Data book)

We can find the LMTD for
Parallel flow.

$$\text{LMTD} = \frac{\Delta T_{in} - \Delta T_{out}}{\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right)}$$

$$\Delta T_{in} = 120 - 30 = 90$$

$$\Delta T_{out} = 85 - 80 = 5$$

$$\text{LMTD} = 29.408^\circ\text{C}$$

$$\textcircled{2} \quad \dot{Q} - \text{Heat Load} = m_c C_{pc} (T_{\text{out}} - T_{\text{in}})$$

$$= 0.8 \times 4.18 (80 - 30)$$

$$= 167.2 \text{ kW}$$

Area required

We have $\dot{Q} = UA \Delta T_{\text{in}}$

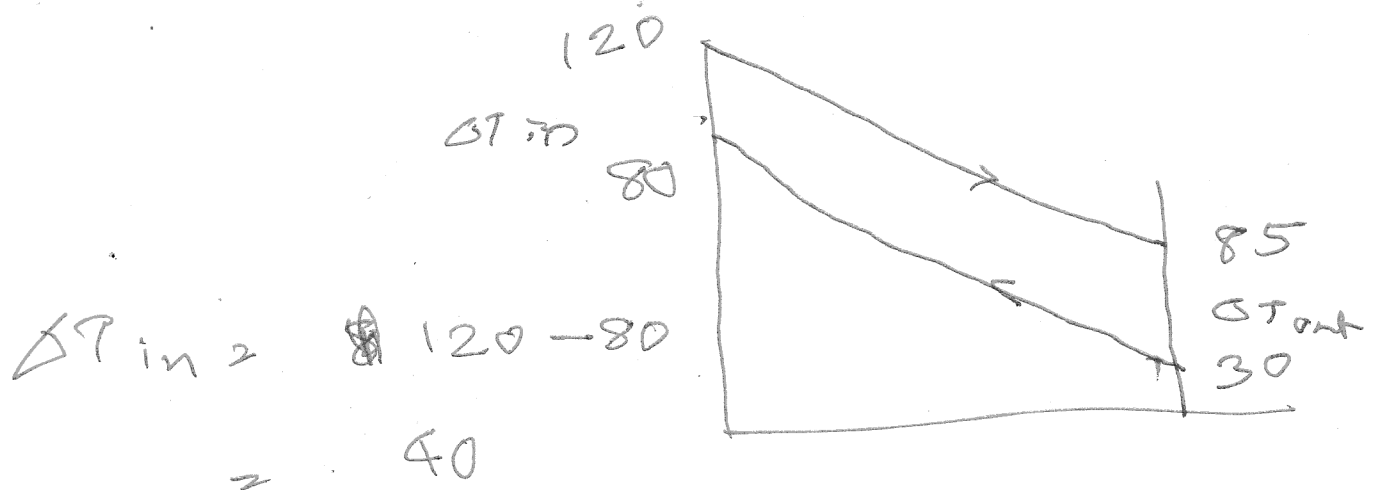
$$A = \frac{\dot{Q}}{U \Delta T_{\text{in}}}$$

$$U = 0.125 \text{ kW/m}^2\text{K}$$

$$A = \frac{167.2}{29.4 \times 0.125}$$

$$= \underline{\underline{45.5 \text{ m}^2}}$$

For Counter Flow



$$LMTD = \frac{40 - 55}{\ln(40/55)}$$

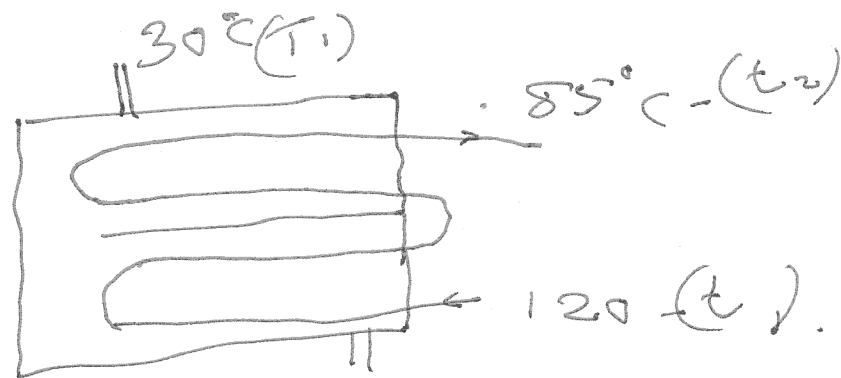
$$= 47.102$$

$$\therefore A = \frac{Q}{U \Delta T_{lm}}$$

$$= \frac{167.2}{0.125 \times 47.102}$$

$$= \underline{\underline{28.4 \text{ m}^2}}$$

For Shell and tube.



$$P = \frac{t_2 - t_1}{T_1 - t_1} = \frac{85 - 120}{30 - 120} = 0.388$$

$$R = \frac{T_1 - T_2}{t_2 - t_1} = \frac{30 - 80}{85 - 120} = 1.43$$

④

Correction factor from

Graph $F = 0.97$

$$Q = UA F \Delta T_{lm, \text{ counterflow}}$$

$$A = \frac{Q}{U F \Delta T_{lm}}$$

$$= \frac{167.2}{0.125 \times 0.97 \times 47.102}$$

$$= \underline{\underline{29.3 \text{ m}^2}}$$

Counter flow heat ex need the least area to meet the same heat load.